Scriptline — AI ideation + scripting pipeline

Python (FastAPI · SQLite · worker pool) + Next.js approval dashboard + Claude API · 2 triggers · 1 human gate · 2 distinct generation stages · 0 manual steps from approval to delivery

main flow · context injected into prompt / rules · learning loop (self-improvement) · AI call · deterministic code

CONTEXT SOURCES

clients/<client_id>/ — a new client is a folder, not a rebuild

reference_pack.md
verbatim client Reference Pack injected into EVERY AI call as the single source of truth (prompt-cached system prefix)

client.yaml
proof bank with IDs (C1…T5)
fact allowlists: numbers, $, %, names
forbidden entities · banned language
CTA terms per funnel stage
avatars · creative & lead types
voice / rhythm / length thresholds

knowledge/craft_playbook.md
client-agnostic craft: hook patterns, PPPC structure, open loops, ICE rubric

prompts/1-4_*.md
4 purpose-built prompts, editable as files

LEARNING MEMORY (SQLite)

the system starts warm on every run

① **Idea memory**
every concept ever generated → never repeated

② **Approver signal**
approve/reject stats by avatar, type, angle + notes

③ **Exemplars**
top-scoring delivered scripts

④ **Watch-outs**
checklist items past drafts failed most

⚡ TRIGGER 1 — start a run
dashboard button · POST /api/runs · CLI · cron

context bus ↓ · ↓ learning bus

AI CALL #1 — GENERATION STAGE 1
STAGE 1 · IDEATION
prompts/1_ideation.md · role: senior creative strategist
batch of 12 concepts, 11 structured fields each
JSON-schema output: proof IDs / avatars / types are enums

pack + playbook · ①②

CODE
Idea guards
validate every field vs client.yaml · compliance scan of the hook
procedural hook on cold traffic → drop · UGC needs a real testimonial
near-duplicate of memory or batch (similarity ≥ 0.62) → drop · 1 top-up

yaml: enums, rules · ①

AI CALL — SEPARATE PROMPT
STAGE 2 · ICE SCORING
prompts/2_ice_scoring.md · role: independent analyst (not the author)
Impact / Confidence / Ease 1–10, written reasoning for each
+ 3-second verdict + risk flags · unscored ideas re-asked

pack + playbook · ②

SQLite · ideas table
11 fields + ICE + reasoning · status = pending · keyed by client_id

all ideas → ①

THE ONE MANUAL STEP
👤 HUMAN APPROVAL GATE — Next.js dashboard
approval queue sorted by ICE · proof text + score reasoning + memory similarity shown
Approve / Reject (+ optional note) — the ONLY manual step

decisions + notes → ②

Approve

⚡ TRIGGER 2 — Approve clicked
POST /api/ideas/{id}/decision

CODE
Atomic handoff (one DB transaction)
status → approved + script row created (UNIQUE per idea): double-clicks
can never spawn two jobs · queued on worker pool · resumed after restart

AI CALL #2 — GENERATION STAGE 2 (SEPARATE PROMPT)
STAGE 5 · SCRIPT GENERATION
prompts/3_scripting.md · role: direct-response scriptwriter
in: full idea record + full text of cited proof + client context
+ exemplars ③ + watch-outs ④ (+ failure report on revision)
out: notes · hook plan (3 interrupt types) · Hook 1/2/3 · Pain › Promise › Proof › CTA
+ open loop + claim ledger: every factual claim → proof ID

pack + yaml CTAs · ③④

yes → rewrite with failure report

STAGE 6 · QUALITY GATE — hybrid; the code decides pass/fail, not the model

CODE — CANNOT BE TALKED OUT OF A VERDICT
Layer A · deterministic checks
#8 every number, $ and % vs. allowlist (copy AND notes) · forbidden entities · invented names
testimonial quotes must be verbatim · claim ledger must cite real IDs
#7 banned language (negation-aware) · no "!" or ALL-CAPS · sentence rhythm
#6 60–180 s for EVERY hook variant · #5 CTA terms per funnel stage
#1 first sentence ≤ 20 words, 3 distinct hooks + interrupt types · #4 proof · #10 sections

yaml: allowlists

hard findings + flags passed in

AI CALL — SEPARATE REVIEWER PROMPT
Layer B · AI reviewer
prompts/4_quality_review.md · scores each of the 10 items 1–5 vs. written anchors
+ evidence quote · 5 voice sub-scores · independent claim audit (ID or UNSUPPORTED)
· rules on context-dependent flags, e.g. "…won't insult you by calling it passive"

pack + checklist

every review's failures → ④

All 10 items pass?
no hard finding AND AI ≥ 4 (+voice ≥ 3, 0 unsupported)

NO

YES

Revisions left?
MAX_REVISIONS = 1

no

⛔ HELD — never delivered
output/<client>/held/NNN_HELD.md failure report
visible in dashboard · "Retry job" button

✅ STAGE 7 · DELIVERY
output/<client>/delivered/NNN_title.docx + .md
p.1 notes (top) · Hook 1/2/3 · main body
p.2 QA scorecard + claim ledger
download from dashboard · status = delivered

delivered → ③ exemplar

Scaling to more clients
new folder in clients/ (pack + yaml) → appears in the dashboard
every DB row keyed by client_id · prompts are templates, no client text
models per stage in .env · SQLite → Postgres is a driver swap

Reliability
SDK retries 429 / 5xx with backoff · invalid JSON re-asked with the error
refusal / truncation → typed error · every step → event log → live UI
crash recovery resumes script jobs · failed jobs retryable from the UI

Gate red-team (built in) + tests
pushes the pack's 3 losing scripts through the real gate on demand
→ all 3 held; report shows which layer caught what
39 automated tests (gate, pipeline, API contract) run offline